

Paper Code: ETEC-208
Paper ID: 28208
Time: 3hrs

Note: Question 1 is compulsory. Internal choice is as indicated.

(2.5 x 10 = 25 marks)
in the strict sense?

Q1. Answer the following questions appropriately

- a) What are stationarity conditions for random processes? When is a random process considered stationary in the strict sense?
b) A certain random variable has the CDF given by:

$$F_x(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ kx^2 & \text{for } 0 < x \leq 10 \\ 100k & \text{for } x > 10 \end{cases}$$

- c) Calculate the value of k for which this definition of CDF is valid. Also find the corresponding probability density function.
d) Discuss the significance of Hilbert transform in context of representation of band pass signals.
e) Comment upon the efficiency of DSB-SC systems with respect to its transmission power.
f) What is the condition that the sideband filters must follow in order for VSB modulation and demodulation to take place?
g) If the bandwidth of DSB-SC and NBFM signals is the same, then explain in what way are they different?

- h) Given an angle modulated signal $x_c(t) = 10 \cos [(10^8)\pi t + 5 \sin(2\pi(10^3)t)]$
Determine the maximum phase deviation and the maximum frequency deviation.
i) What is image frequency and what is the problem it causes in signal reception? How does superheterodyne receiver take care of this problem?
j) A receiver is connected to an antenna whose resistance is 50Ω has an equivalent noise resistance of 30Ω . Calculate the receiver's Noise figure and its equivalent noise temperature.
k) With respect to noise performance, which is the better modulation scheme – DSB-SC or DSB-AM? Why? (3 marks)

UNIT – I

- Q2. a) If two random variables are said to be statistically independent, what does this imply? (3 marks)
b) Define the term 'Variance' for the random variable X. Calculate the same if PDF for X is given by, (7 marks)

$$f_x(x) = \frac{1}{2} e^{-|x|} \text{ for } -\infty < x < \infty$$

- c) What are the typical properties of probability density function and cumulative distribution functions for random variables? (2.5 marks)

- Q3. a) Write a detailed note on Gaussian random process. Discuss why a Gaussian random process is said to be one of the most commonly occurring natural processes. (5 marks)
b) Let x and y be independent random variables with variance σ_1^2 and σ_2^2 respectively. Consider the sum, $z = ax + (1-a)y$ $0 \leq a \leq 1$ (7.5 marks)
Find the value of 'a' that minimizes the variance of z.

UNIT – II

- Q4. a) Discuss the need of modulation with respect to analog communication systems. (5 marks)
b) What is the problem that is encountered in case there is loss in coherence while demodulating an AM signal using a product demodulator? (5 marks)
c) Prove that for 100% modulated DSB-AM signal, the power efficiency is 33.33%. (2.5 marks)

OR

- Q5. a) What is meant by harmonic distortion in context of analog communication systems? (3 marks)
b) Discuss AM demodulation using envelop detector. Also discuss the peak clipping effect. (6 marks)
c) Show that a synchronous (coherent) demodulator can demodulate an AM signal $x_{AM}(t) = [A + m(t)] \cos \omega_c t$ (3.5 marks)
regardless of the value of A.

UNIT – III

- Q6. a) Discuss Varactor diode based direct FM generation technique. (5 marks)
b) Describe the operation of a super-heterodyne receiver with the help of a block diagram. Explain the precise relevance of each block used in the receiver structure. (7.5 marks)

OR

- Q7. a) Write a detailed note on use of PLL for FM demodulation. (6.5 marks)
b) Write a short note on the following (3 marks)
1. AGC
2. Frequency mixers (3 marks)

UNIT – IV

- Q8. a) Discuss the various sources of noise in communication systems. (5 marks)
b) Prove that the performance of an SSB systems using synchronous detection is equivalent to the performance of both DSB and baseband systems. (7.5 marks)

OR

- Q9. a) The first stage of a two stage amplifier has a voltage gain of 10, a 600Ω resistor, a 1600Ω equivalent noise resistance and a $27k\Omega$ output resistor. For the second stage, these values are 25, $81k\Omega$, $10k\Omega$ and $1M\Omega$ respectively. Calculate (5 marks)
1. Equivalent input noise resistance of this two stage amplifier (4.5 marks)
2. Noise figure of the amplifier driven by a generator with input impedance of 50Ω . (3 marks)
b) What is the significance of pre-emphasis and de-emphasis filtering in context of noise performance of FM systems. (5 marks)

(Please write your Exam Roll No.)

Exam Roll No.

FIRST TERM EXAMINATION

FOURTH SEMESTER [B.TECH] FEBRUARY 2015

Paper Code: ETEC-208 (ECE)

Subject: Communication Systems

Time: 1 Hour

Maximum Marks: 30

Note: Attempt three questions including Q. no. 1 which is compulsory.

Q1

(2.5x4 = 10)

- (a) Why are over modulated and suppressed carrier AM signals a problem from the receiver point of view? In what situations might DSB-SC be preferred to conventional AM?
- (b) Show that a signal and its Hilbert transform have the same energy content.
- (c) What is Ergodic Process? Explain.
- (d) The joint PDF of X and Y is given by $p_{X,Y}(x,y) = xy e^{-(x^2+y^2)/2} u(x)u(y)$. Find the marginal probability density functions $p_X(x)$ and $p_Y(y)$.

Q2

(5x2 = 10)

- (a) Consider the probability density $p_X(x) = ae^{-b|x|}$, where X is a random variable whose allowable values range from $x = -\infty$ to $x = +\infty$. Find (a) The cumulative distribution function $F_X(x)$, (b) The relationship between a and b, & (c) The probability that the outcome X lies between 1 and 2.
- (b) The baseband signal $m(t)$ in the frequency - translated signal $v(t) = m(t) \cos 2\pi f_c t$ is recovered by multiplying $v(t)$ by the waveform $\cos 2\pi(f_c + \Delta f)t$. The product waveform is transmitted through a low pass filter which rejects the double- frequency signal. Find the output signal of the filter.

Q3

(5x2 = 10)

- (a) Explain the working of envelope detector with input and output waveform and also write the importance of RC time constants for this detector.
- (b) Write a short notes on communication channels and their mathematical models.

Q4

(5x2 = 10)

- (a) Consider the AM signal $s_{AM}(t) = [A_c + m(t)] \cos 5000t$ where the modulating signal is Given by $m(t) = 3\cos 50t + 5\cos 150t$. Let the modulation index be 0.8. Find;
 - (i) The amplitude of carrier, A_c ,
 - (ii) The carrier power, and
 - (iii) The transmission efficiency
- (b) Consider a random process $X(t)$ is given by $X(t) = A \cos (w_0 t + \phi)$. A and w are deterministic and $\phi \sim U[-\pi, +\pi]$. Verify that $X(t)$ is WSS.

I Sessional.

Q1. @ Over modulated and suppressed carrier AM signals are impossible to recover using simple envelope detection. The ~~env~~ envelope of the mixed output i.e. product of the low frequency message and high frequency carrier, is proportional to ~~the~~ $|m(t)|$ where $m(t)$ is the message signal. Thus the output of the envelope detector, ~~again~~ will give $|m(t)|$ instead of the ~~the~~ original message. To avoid this problem, one may use a product demodulator or a coherent detector. This system requires the incoming modulated signal to be multiplied with the carrier frequency wave with exact phase matching. This exact frequency and phase matching is very difficult to achieve and adds to the complexity as well as the cost of the receiver.

DSB-SC system may be preferred in situations where power of transmission needs to be limited. DSB-AM needs a high transmission power but has a simple demodulator. This makes DSB-AM suitable for broadcast communication systems. DSB-SC, however, requires a more complicated receiver but has small power of transmission. One such application is telephony.

(b) Hilbert transform can be defined as

$$\hat{H}(t) = -j \operatorname{sgn}(t) H(t)$$

Energy of Hilbert transformed function $\hat{H}(t)$ can be written as

$$\hat{E} = \int_{-\infty}^{\infty} |\hat{H}(t)|^2 dt$$

$$= \int_{-\infty}^{\infty} |-j \operatorname{sgn}(t) H(t)|^2 dt$$

$$= \int_{-\infty}^{\infty} |H(t)|^2 dt$$

$$\therefore \text{as } |-j \operatorname{sgn}(t) H(t)| = \begin{cases} |H(t)| & t \geq 0 \\ |H(t)| & t < 0 \end{cases}$$
$$= |H(t)| \text{ for all } t$$

$$\Rightarrow \boxed{\hat{E} = E}$$

Hence energy content of the signal and its Hilbert transform are the same.

Hilbert transform

1) Ergodic Process

A random process is ergodic if its statistical averages can be replaced with (i.e. equal to) its time averages. ^{therefore simply the} statistical averages are ~~replaced~~ time averages for any sample functions.

In many practical systems, we may not have the knowledge of the distribution function of the Random Process. Then for the single sample function that we may observe, how can we evaluate the mean, or other moments? In such a case if the process is ergodic, then the time averages of a single sample function can be treated as statistical averages. Mathematically, time averages can be evaluated as

$$\bar{x}(t) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt$$

Hence for an ergodic process,

$$E(x) = \int_{-\infty}^{\infty} x p_x(x) dx = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt$$

As time averages can never be time dependent, thus all possible moments and means of the ergodic process are time independent. Hence all ergodic processes are strict sense stationary.

(d)

$$p_{xy}(x, y) = xy e^{-(x^2+y^2)/2} u(x) u(y)$$

$$p_x(x) = \int_{-\infty}^{\infty} p_{xy}(x, y) dy$$

$$= \int_{-\infty}^{\infty} xy e^{-(x^2+y^2)/2} u(x) u(y) dy$$

$$= \int_{-\infty}^{\infty} xy e^{-(x^2/2)} \cdot e^{-(y^2/2)} dy$$

$$= x \cdot e^{-x^2/2} u(x) \int_{-\infty}^{\infty} y e^{-y^2/2} dy$$

Assume $y^2 = t \Rightarrow y dy = dt$
 $\frac{y^2}{2} = t$

$$p_x(x) = x \cdot e^{-x^2/2} \cdot u(x)$$

Similarly $p_y(y) = \int_{-\infty}^{\infty} p_{xy}(x, y) dx$

$$= y \cdot e^{-y^2/2} u(y) \int_{-\infty}^{\infty} x e^{-x^2/2} u(x) dx$$

$$p_y(y) = y \cdot e^{-y^2/2} \cdot u(y)$$

Hence, $p_x(x)$ and $p_y(y)$ are the required marginalised ~~density~~ ^{density} functions.

$$2(a) \quad p_x(x) = a \cdot e^{-b|x|}$$

As probability can never be negative, hence $a > 0$. Also to maintain the probability below 1, b is also always positive.

$$\Rightarrow p_x(x) = \begin{cases} a e^{-bx} & x \geq 0 \\ a e^{bx} & x < 0 \end{cases}$$

$\Rightarrow$ Distribution function

$$P_x(x) = \begin{cases} \int_{-\infty}^x a e^{bx'} dx' & x < 0 \\ \int_{-\infty}^0 a e^{bx'} dx' + \int_0^x a e^{-bx'} dx' & x > 0 \end{cases}$$

$$\int_{-\infty}^x a \cdot e^{bx'} dx' = \frac{a}{b} [e^{bx'}]$$

$$\int_{-\infty}^0 a e^{bx'} dx' = \frac{a}{b}$$

$$\int_0^x a e^{-bx'} dx' = \frac{a}{b} [1 - e^{-bx}]$$

$$\Rightarrow P_x(x) = \begin{cases} \frac{a}{b} e^{bx} & x < 0 \\ \frac{a}{b} [2 - e^{-bx}] & x \geq 0 \end{cases} \rightarrow (1)$$

$$\text{At } x = \infty, P_x(x) = 1$$

$$\Rightarrow \frac{a}{b} [2 - 0] = 1 \Rightarrow \boxed{2a = b} \rightarrow (2)$$

$$\boxed{P(1 \leq x \leq 2) = P_x(2) - P_x(1) = \frac{a}{b} [e^{-b} - e^{-2b}]} \rightarrow (3)$$

(b) Let the product waveform be written as

$$x(t) = m(t) \cos(2\pi f_c t) \cdot \cos(2\pi(f_c + \Delta f)t)$$

$$= \frac{m(t)}{2} [\cos(4\pi f_c t + 2\pi \Delta f t) + \cos(2\pi \Delta f t)]$$

As the low pass filter rejects the double frequency signal the output we are left with $x'(t)$ is

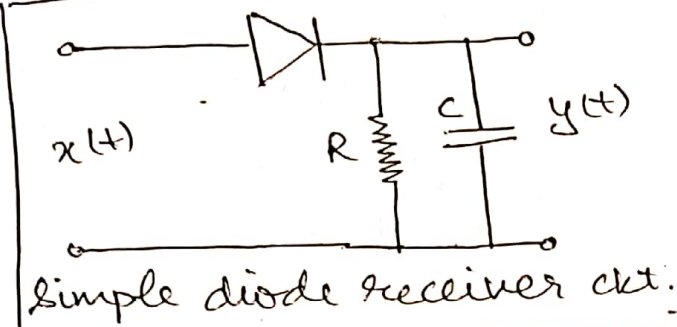
$$x'(t) = \frac{m(t)}{2} \cos(2\pi \Delta f t)$$

Here we see that if Δf is a small variation, then the output is some kind of a frequency translated signal considering $m(t)$ as the carrier. This highlights the fact that frequency matching is imperative for product demodulation.

~~Note here that DSB-AM can also be demodulated by coherent receiver but such a practice is generally avoided.~~

Q3 AM demodulation / Diode Detector / Envelop detector

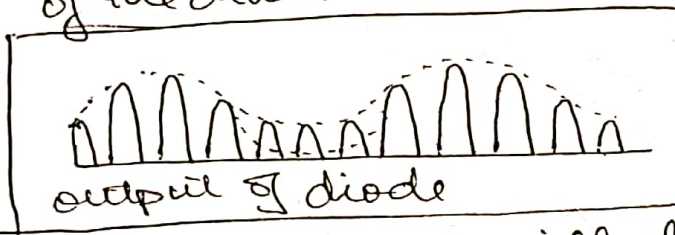
(a)



*If a simple diode receiver gives an output $y(t)$ that is proportional to message signal $m(t)$, no synchronous carrier is required. Thus, expensive carrier recovery ckt. is also not needed.

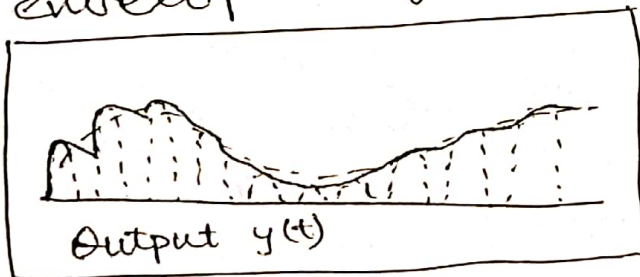
Working

Consider the waveform of $x(t)$. Here diode will be as a rectifier and hence the positive half cycle (only) will be able to conduct the diode. Therefore the output of the diode can be drawn as follows:



Assuming a large resistance R , the entire output will attempt to charge the capacitor C . Charging will happen quickly

and the output will almost trace back the incoming pulse of positive voltage. When the pulse begins to fall, the capacitor starts to discharge through R but that process is slow. As charging starts again quite quickly, the output waveform roughly traces the envelop waveform as shown below.



This output can further be smoothed using an appropriate LPF. Let us examine the condition under which this circuit functions properly.

Firstly, let us look at the charging path of the capacitor. Assume a source resistance R_s and a forward biased diode resistance r_f . Then the charging path will be a series combination of R_s & r_f . The discharging path will be through load R .

for appropriate working of this circuit, the charging time constant must be extremely small in comparison to the discharging time constant.

$$\Rightarrow RC \gg (R_s + r_f)C$$

$$\Rightarrow R \gg (R_s + r_f)$$

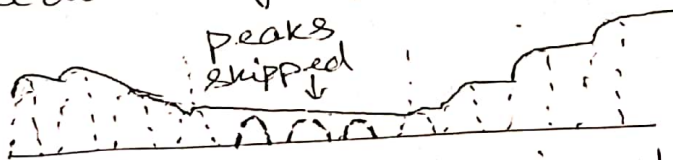
Another way to look at it is that charging should occur much faster in comparison to the time period of the carrier waveform.

$$\Rightarrow (R_s + r_f)C \ll \frac{1}{f_c}$$

Thus charging starts as soon as positive slope is encountered in the true half cycle. Similarly we can observe that discharging time constant must be very large in comparison to carrier time period.

$$\Rightarrow RC \gg \frac{1}{f_c}$$

But what may occur here is that RC might be so large that some smaller peaks may be missed while discharging as shown below:



This results in a distortion which is known as clipping distortion. Thus RC must be larger than carrier time period, but must be smaller than the maximum time period of the message signal.

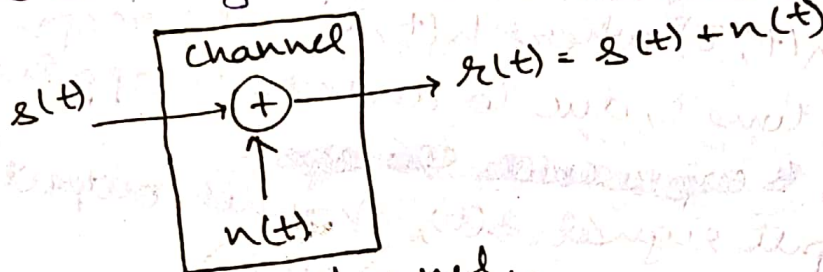
$$\Rightarrow \frac{1}{B} \gg RC \gg \frac{1}{f_c}$$

Broadly, communication models can be categorized into three broad mathematical models.

(a) The Additive Noise Channel

In this model the transmitted signal $s(t)$ is corrupted by an additive random noise process $n(t)$. It is assumed however, that the signal remains unchanged itself.

Physically, the additive noise process may arise from electronic components and amplifiers at the receiver of the communication system, or from interference encountered in transmission, as in the case of radio signal transmission.

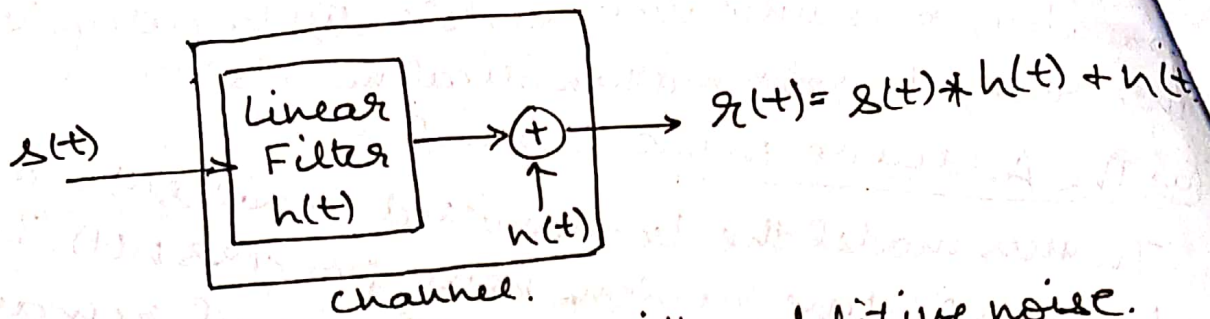


Additive Noise Channel.

(b) Linear filter channel

In some physical channels such as wireline telephone channels, filters are used to ensure that the transmitted signals do not exceed specified bandwidth limitations and thus do not interfere with one another. Such channels are generally characterised mathematically as linear filter channels with additive noise. Hence, if the channel input is the signal $s(t)$, the channel output is the signal.

$$\begin{aligned} r(t) &= s(t) * h(t) + n(t) \\ &= \int_{-\infty}^{\infty} h(\tau) s(t-\tau) d\tau + n(t). \end{aligned}$$



Linear Filter channel with additive noise.

(c) Linear Time Variant Filter Channel

Physical channels such as underwater acoustic channels and ionospheric radio channels which result in time-variant multipath propagation of the transmitted signal may be characterized mathematically as time-variant channels impulse response $h(t; \tau)$ where $h(t; \tau)$ is the response of the channel at time t , due to an impulse applied at time $t - \tau$.

For an input signal $s(t)$, channel output signal is

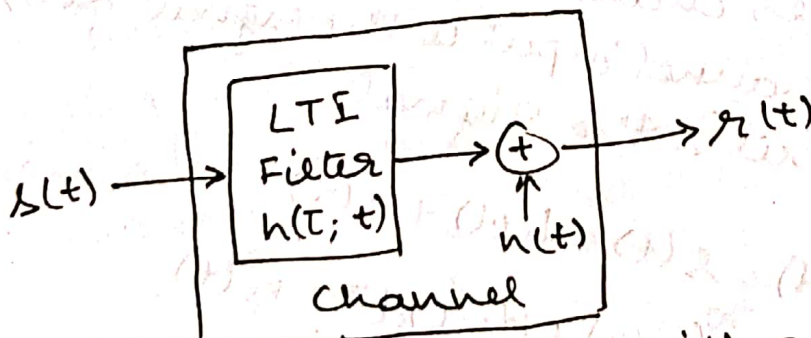
$$r(t) = s(t) * h(t; t) + n(t)$$

$$= \int_{-\infty}^{\infty} h(\tau; t) s(t - \tau) d\tau + n(t)$$

$$\text{where } h(\tau; t) = \sum_{k=1}^L a_k(t) \delta(\tau - \tau_k)$$

$\Rightarrow r(t)$ can be re-written as

$$r(t) = \sum_{k=1}^L a_k(t) s(t - \tau_k) + n(t)$$



LTI Filter channel with additive noise.

$$1(a) \quad s_{AM}(t) = [A_c + m(t)] \cos 5000t \dots$$

$$= A_c \left[1 + \frac{m(t)}{A_c} \right] \cos 5000t.$$

$$m(t) = 3 \cos 50t + 5 \cos 150t.$$

Here ^{min}~~max~~ value of $m(t)$ will be

$$m(t)_{\min} = -8$$

$$\text{Minimum value of } \frac{m(t)}{A_c} = \frac{-8}{A_c}$$

$$\text{Similarly Max value of } \frac{m(t)}{A_c} = \frac{8}{A_c}$$

As modulation index is $\mu = 0.8$

$$0.8 = \frac{\frac{8}{A_c} - \frac{-8}{A_c}}{2 + \frac{8}{A_c} + \frac{-8}{A_c}}$$

$$= \frac{16/A_c}{2}$$

$$= \frac{8}{A_c}$$

$$\Rightarrow \boxed{A_c = 10} \quad \text{--- (1)}$$

Carrier Power = $\frac{A_c^2}{2}$ as carrier is sinusoidal

$$\boxed{S_c = P_c = 50 \text{ W}} \quad \text{--- (2)}$$

Signal power being transmitted is scaled down by A_c .

$$\Rightarrow \text{As } \frac{m(t)}{A_c} = 0.3 \cos 50t + 0.5 \cos 150t.$$

$$\text{Power of } \frac{m(t)}{A_c} = \frac{0.09}{2} + \frac{0.25}{2}$$

$$S_m = 0.17.$$

$$\text{Transmission efficiency} = \frac{S_m}{1 + S_m} \times 100$$

$$= \frac{0.17}{1.17} \times 100$$

$$\boxed{E = 14.53\%}$$

(6) Considering uniform distribution of ϕ , its density function is given by

$$P_{\phi}(\phi) = \begin{cases} \frac{1}{2\pi} & -\pi \leq \phi \leq \pi \\ 0 & \text{otherwise.} \end{cases}$$

Mean value of $x(t)$

$$E[x(t)] = \int_{-\infty}^{\infty} x(t) P_{\phi}(\phi) d\phi$$

$$= \int_{-\pi}^{\pi} \frac{A}{2\pi} \cos(\omega_c t + \phi) d\phi$$

Over a single period, cosine wave integrates to zero

$$\Rightarrow E[x(t)] = 0$$

mean squared value

$$E[x^2(t)] = \int_{-\infty}^{\infty} \frac{A^2}{2\pi} \cos^2(\omega_0 t + \phi) d\phi.$$

$$= \frac{A^2}{4\pi} \int_{-\pi}^{\pi} (1 + \cos(2\omega_0 t + 2\phi)) d\phi$$

$$= \frac{A^2}{4\pi} [2\pi]$$

$$= \frac{A^2}{2}$$

Autocorrelation

$$R_x(\tau) = E[x(t)x(t+\tau)]$$

$$= \int_{-\pi}^{\pi} \frac{A^2}{2\pi} \cos(\omega_0 t + \phi) \cos(\omega_0(t+\tau) + \phi) d\phi$$

$$= \frac{A^2}{4\pi} \left[\int_{-\pi}^{\pi} \cos(2\omega_0 t + 2\phi + \omega_0 \tau) d\phi + \int_{-\pi}^{\pi} \cos \omega_0(\tau) d\phi \right]$$

$$= \frac{A^2}{4\pi} \cdot 2\pi \cdot \cos \omega_0 \tau$$

$$= \frac{A^2}{2} \cos \omega_0 \tau$$

Since the first order and second order means are time independent and autocorrelation is a function only in the time difference, this process is wide sense stationary.

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SECOND TERM EXAMINATION

FOURTH SEMESTER [B.TECH] ~~FEBRUARY~~ ^{APRIL} 2015

Paper Code: ETEC-208 (ECE)

Subject: Communication Systems

Time: 1 Hour

Maximum Marks: 30

Note: Attempt three questions including Q. no. 1 which is compulsory.

Q1 (2.5x4 = 10)

- Why FM is called constant bandwidth system?
- Compute the bandwidth requirement for the transmission of an FM signal having a frequency deviation of 75 kHz and audio frequency of 10 kHz.
- A superheterodyne AM broadcast receiver has an IF of 455 kHz and it is tuned to a frequency of 700 kHz, determine the image frequency.
- Distinguish between narrow band FM and wide band FM.

Q2 (5x2 = 10)

- Describe basic reactance modulator used to generate FM signal.
- Discuss the advantages of superheterodyne receiver over the TRF receiver. Explain how the constant intermediate frequency is achieved in the superheterodyne receiver?

OR

An FM wave is represented by $s(t) = 10 \cos(6\pi \times 10^3 t + 6 \sin 1250\pi t)$. Find

- The carrier and modulating frequencies.
- The modulation index.
- Is it narrow band or wideband FM?
- Maximum deviation of FM wave.
- What power this FM will dissipate in a 10 ohm resistor?

Q3 (5x2 = 10)

- A 25 MHz carrier is modulated by a 40 Hz audio sine wave. If the carrier voltage is 4V and the maximum deviation is 10 kHz, write the equation of this modulated wave for
 - FM and
 - PM.
- Explain the phase-shift method of SSB generation with the help of neat block diagram.

Q4 (5x2 = 10)

- Explain Armstrong frequency modulation system with block diagram.
- What is pre-emphasis and de-emphasis? Why is it used?

Second Sessional Solutions

① @ Why is FM called constant bandwidth system?

Ans Consider a general FM system with wide band spectrum. In such a case, the frequency deviation (maximum) is large in comparison to the input signal frequency. Thus the bandwidth, by Carson's rule, i.e. $B.W = 2(\Delta f + f_m)$, will not change much by change in the incoming message signal frequencies. Thus, it is sometimes referred to as constant bandwidth system.

② Compute the bandwidth requirement for the transmission of an FM signal having a frequency deviation of 75 kHz and audio frequency of 10 kHz?

Ans Bandwidth of FM signal is given by Carson's rule

$$\begin{aligned} B.W &= 2(\Delta f + f_m) \\ &= 2(75 + 10) \text{ kHz} \\ &= 170 \text{ kHz} \end{aligned}$$

$\therefore$ Bandwidth of FM signal is 170 kHz.

③ A superheterodyne AM broadcast receiver has an IF of 455 kHz and it is tuned to a frequency of 700 kHz, determine the image frequency.

Ans Carrier frequency = 700 kHz.
Intermediate frequency = 455 kHz.
Mixing system = Superheterodyne
i.e. $f_c + f_{IF}$.

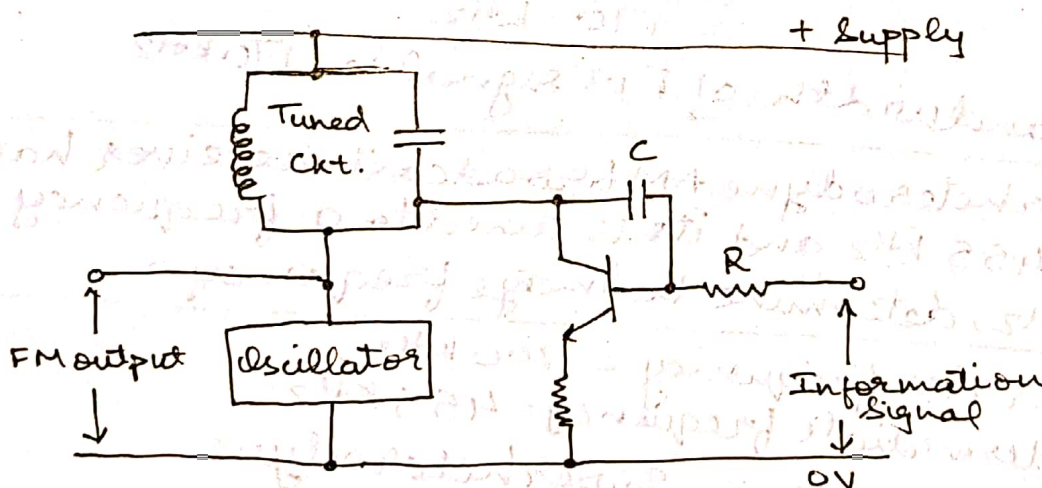
$$\begin{aligned} \Rightarrow \text{Image frequency} &= f_c + 2f_{IF} \\ &= 700 + 2(455) \\ &= 1610 \text{ kHz} = 1.61 \text{ MHz} \end{aligned}$$

(d) Distinguish between Narrow band FM and Wide band FM.

Ans	Narrowband FM	Wideband FM
1.	Modulation index $\beta = \frac{k_f A_m}{f_m}$ is small in comparison to 1.	Modulation index $\beta = \frac{k_f A_m}{f_m}$ is large in comparison to 1.
2.	Bandwidth is nearly equal to $2f_m$.	Bandwidth is nearly equal to $2(f_m + \Delta f_{max})$
3.	Generation of Narrow band FM uses DSB-SC modulator. Direct FM generation techniques often fail in case of Narrowband FM	Wideband FM can be generated directly using VCO technique. Even Narrowband FM can be converted to Wideband FM using Armstrong Method.

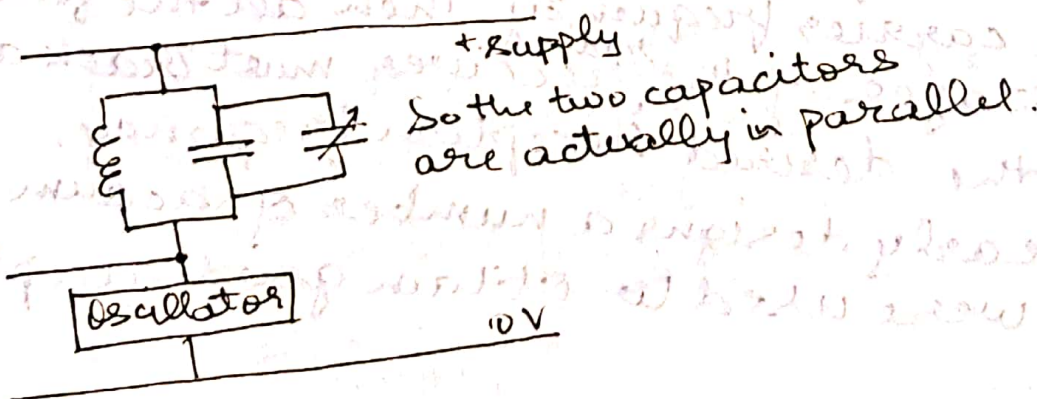
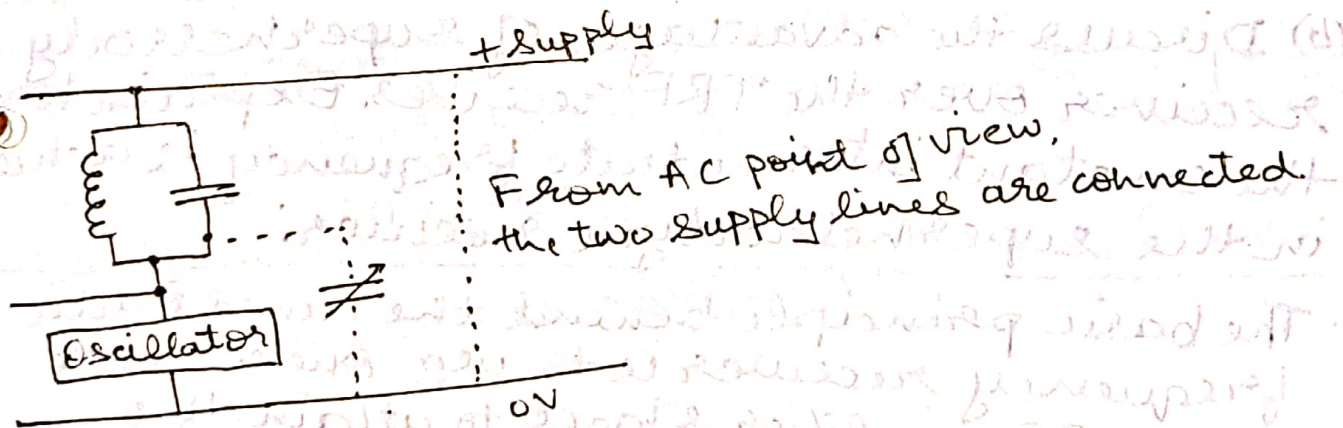
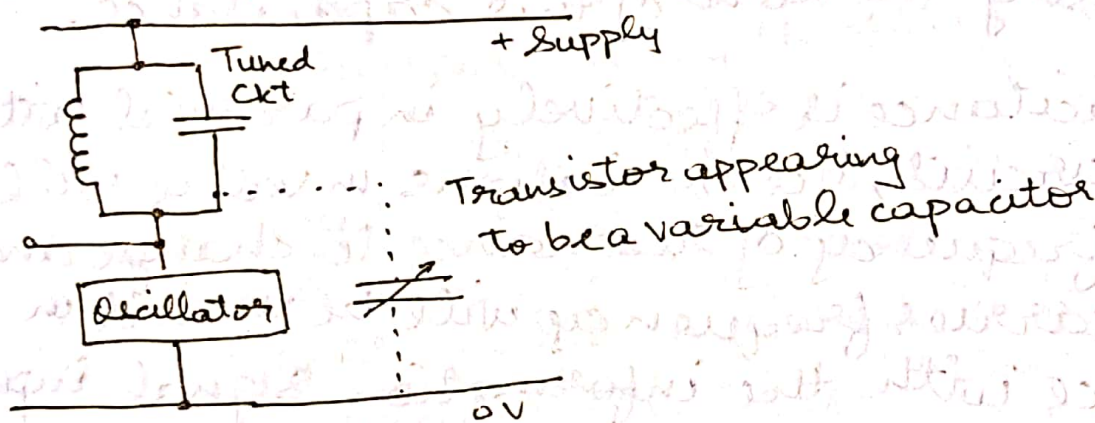
Q2 (a) Describe basic reactance modulator used to generate FM signal.

Ans The block diagram shown below shows a reactance modulator.



In the figure, the tuned circuit and oscillator generate the unmodulated carrier. The capacitor C and the resistor R are the two components used for the phase shifting, and together with the transistor, form the voltage controlled capacitor. This voltage-controlled capacitor is actually in parallel with the tuned circuit.

This can be justified by the observation that from an AC point of view, the two power supplies are connected. This process can be summed up in the following diagrams.



Operation of the Reactance modulator

1. The oscillator and tuned circuit provide the unmodulated carrier frequency and this frequency is present on the collector of the transistor.
2. The capacitor and the resistor provide the 90° phase shift between the collector voltage and current. This ~~is~~ makes the circuit appear as a capacitor.
3. The changing information signal being applied to the base has the same effect as changing the bias voltage applied to the transistor and this would have the effect of increasing and decreasing the value of this capacitance.

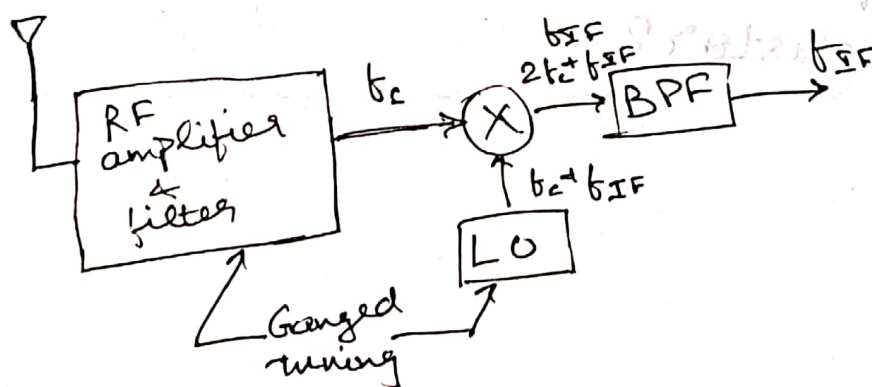
As the capacitance is effectively in parallel with the tuned circuit, the variations in value will cause the frequency of resonance to change and hence the carrier frequency will be varied in accordance with the information signal input.

Q2(b) Discuss the advantages of superhetrodyne receiver over the TRF receiver. Explain how the constant intermediate frequency is achieved in the superhetrodyne receiver.

- Ans. The basic principle behind the Tuned Radio frequency receiver is to use one or more tuned RF amplifier stages to attain the desired carrier frequency. Then all the following tuned stages of the ^{radio} receiver must track and tune to the desired reception frequency. In the early designs a number of vacuum tubes were used to obtain good reception.

The disadvantage of this process however is the poor selectivity and low sensitivity in proportion to the number of tubes employed. Selectivity requires narrow bandwidth for the tuned RF amplifier but that is only possible with high Q filters or many filter sections. This problem extends to all tuning components of the receiver. In contrast, the problem of achieving constant sensitivity and bandwidth over a range of frequencies arises only in one circuit in the Superhetrodyne receiver; in the first stage i.e. RF amplifier.

The complicated tuning issues of the TRF receiver are also taken care of by the intermediate frequency heterodyning. Thus tuning is only required to be done precisely for one stage that tunes the RF amplifier to f_c and the local oscillator to $f_c + f_{IF}$. After the heterodyning the central frequency remains the same at f_{IF} and thus circuitry based on that frequency can be easily designed. The heterodyning stage can be explained as follows.



The tuning ckt tunes the RF amplifier and ~~and~~ to the required carrier frequency. This stage is highly selective with required bandwidth and thus allows only the required signal to ~~the~~ be amplified and others to be shunned. The LO i.e. Local oscillator is set to be tuned parallelly to $f_c + f_{IF}$ using ganged tuning. The mixer then ~~&~~ mixes f_c and $f_c + f_{IF}$ to generate the frequencies $|f_c \pm (f_c + f_{IF})|$, i.e. f_{IF} and $2f_c + f_{IF}$. A following band pass filter tuned at f_{IF} blocks the high frequency component and permits the ~~the~~ modulated signal shifted to f_{IF} to be sent to the detector to recover the original message signal.

Q2(b) An FM wave is represented by

OR part

$$s(t) = 10 \cos(6\pi \times 10^6 t + 6 \sin 1250 \pi t)$$

Find

- (1) The carrier and modulating frequencies
- (2) The modulation index
- (3) Is it narrowband or wideband
- (4) Maximum deviation of FM wave
- (5) What power this FM will dissipate in a 10Ω resistor?

General equation of FM signal is

$$s(t) = A_c \cos \left(2\pi f_c t + k_f A_m \int_{-\infty}^t \cos(2\pi f_m \tau) d\tau \right) \\ = A_c \cos \left(2\pi f_c t + \frac{k_f A_m}{2\pi f_m} \sin 2\pi f_m t \right)$$

Comparing this with the given FM signal

① Carrier frequency $f_c = \frac{6\pi \times 10^6}{2\pi}$

$$f_c = 3 \times 10^6 \text{ Hz}$$

② Signal frequency $f_m = \frac{1250\pi}{2\pi} = 625 \text{ Hz}$

③ Modulation index, $\beta = \frac{k_f A_m}{2\pi f_m} = 6$

③ As $\beta > 1$, this is a wideband FM system

④ Maximum deviation is given by

$$\Delta f = \beta f_m = 6 \times 625 \\ = 3750 \text{ Hz}$$

⑤ As the amplitude of the FM signal remains constant, the power dissipated can be evaluated as

$$P = \frac{V_{rms}^2}{R} = \frac{(10/\sqrt{2})^2}{10} \\ = 5 \text{ W}$$

Q3@ A 25 MHz carrier is modulated by a 40 Hz audio sine wave. If the carrier voltage is 4V and the maximum deviation is 10 kHz, write the equation of this modulated wave for

- (1) FM
- (2) PM.

Ans. Given $A_c = 4V$
 $f_c = 25 \times 10^6 \text{ Hz}$
 $f_m = 40 \text{ Hz}$
 $\Delta f_{\max} = K_f A_m = 10 \times 10^3 \text{ Hz}$

$\therefore$ FM signal given by
 $x_{FM}(t) = A_c \cos \left(2\pi f_c t + K_f A_m \int_{-\infty}^t \cos 2\pi f_m \tau d\tau \right)$
 $\Rightarrow x_{FM}(t) = 4 \cos \left(50\pi \times 10^6 t + \frac{2\pi \times 10 \times 10^3}{2\pi \times 40} \sin 80\pi t \right)$

or $x_{FM}(t) = 4 \cos \left(50\pi \times 10^6 t + 250 \sin 80\pi t \right)$

For the same deviation, the PM signal is given by

$x_{PM}(t) = A_c \cos \left(2\pi f_c t + K_p A_m \cos 2\pi f_m t \right)$
 $\Rightarrow x_{PM}(t) = 4 \cos \left(50\pi \times 10^6 t + 10^4 \cos 80\pi t \right)$

Q3(b) Phasing Method of SSB as given in notes

Q4(a) Armstrong Method from notes

Q4(b) Pre-emphasis, De-emphasis as given in notes